



اَوْنُوْرَسِيَّتِي تِيَكْنُوْلُوْجِي مَبَارَا
UNIVERSITI
TEKNOLOGI
MARA

**FACULTY OF INFORMATION SCIENCE UNIVERSITI TEKNOLOGI MARA PUNCAK
PERDANA CAMPUS**

ADVANCED WEB DESIGN DEVELOPMENT AND CONTENT MANAGEMENT (IMS 566)

FINAL GROUP PROJECT

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Project Background

In higher education institutions, student frequently need to be engaged to finish their assignment. Some of the subject require some interaction with organization or companies for activities such as academic interviews and field research to collect the information. Officially requesting these activities requires formal university formal authorization which is typically issued via an official letter known as “Surat Sokongan”. Historically managing, requesting and verifying this letter has been a manual or decentralized process, causing delays and administrative bottleneck for both student and university administrators.

The SURAT Application is a web-based system developed to digitize, streamline and centralize the entire process of official university letter approval. This will help student to view whether their application has been approved or reject. Meanwhile it also reduces the delay between student and university administrators after the application has been send. By transitioning these manual workflows into a secure, centralized digital environment the SURAT application became a bridge to connect gap between student body and university administration, fostering a more transparent and agile academic ecosystem.

Problem Statement

Before the implementation of the SURAT system, the process of requesting official interview or reference letters faced several key challenges:

- **Inefficient Workflows:** Students had to manually submit company and group details, leading to communication delays.
- **Administrative Burden:** University administrators had to manually review student requests, draft individual letters, and verify details, consuming valuable time.
- **Lack of Centralization:** Tracking the status of an application (whether it was pending, approved, or rejected) was difficult for students, leading to uncertainty and lost academic opportunities.

Proposed Solution

The SURAT system acts as a bridge between students and university management. The system operates through an automated workflow consisting of three main phases:

1. **Application Submission:** Students input comprehensive details regarding their academic program, faculty (such as the *Faculty of Computer and Mathematical Sciences / CDIM*), specific course codes (e.g., *CDIM262*), target company information, and student group details.
2. **Administrative Review:** University administrators access an exclusive dashboard to inspect the submitted data. They can immediately approve or reject the request based on the validity of the provided details.
3. **Automated Letter Generation:** Once approved, the system changes the status to "Approved" and automatically generates an official, dynamic PDF letter featuring the university logo and a secure validation QR code, which students can instantly download.

Technical Architecture

The application is built using a reliable open-source web tech stack to ensure data integrity and smooth processing:

- **Backend Framework:** Developed using CakePHP, utilizing MVC architecture to handle business logic, data routing, and secure authentication.
- **Database Management:** Managed through phpMyAdmin (MySQL) to store user credentials, student group records, application tracking metrics, and company details.
- **Core Functionality:** Integrated with dynamic PDF generation libraries to format official institutional letters automatically upon admin approval.

Inside The SURAT Application

# User	Company Name	Group Leader Name	Date Interview	Method	status	Actions
1 Aiman ayer SARJANA MUDA SAINS MAKLUMAT (KEPUJIAN) PENGURUSAN SISTEM MAKLUMAT FACULTY SCIENCE INFORMATION	Abc Sdn.Bhd Jalan Utapo Jalan 11/11 70400 Nilai Negeri Sembilan	aiman ayer 0112223333 aiman@ayer.com 4	Jun 08,26	Online	Approved	View Edit Delete
2 Aiman NurHakimin SARJANA MUDA SAINS MAKLUMAT (KEPUJIAN) PENGURUSAN SISTEM MAKLUMAT FACULTY SCIENCE INFORMATION	Dola Sdn.Bhd No. 15, Jalan Teknologi 3/1 Kota Damansara 47810 Petaling Jaya Selangor	Aiman NurHakimin 0112223333 aiman@ayer.com 4	Jun 29,26	Physical	Approved	View Edit Delete

< Previous Next >

Page 1 of 1, showing 2 record(s) out of 2 total

Picture 1.

The system manages and keeps track of official students' interview requests on the "Application List" interface using a structured data grid. All the applicant information such as the applicant's name, faculty and academic specialization of the student and the name and full address of the target company are displayed in each entry on this page. In addition, key logistical data are captured, like group leader contact details, number of group members, interview date, and the type of contact via online or physical with the group. Here administrators can easily see the progress of each request using a visual status badge indicating success as "Approved" in green and using these top navigation tabs to gain access to lists, reports, and export to external archive. For direct record management, a specific group of action utilities is provided on the right that enables the administrator to check out deep files, edit record details or even remove records from the system database.



Ref Number : 12345
 Tarikh: : 01 January 1970

Abc Sdn.Bhd
 jalan Utapo
 jalan 11/11
 70400
 Nilai
 Negeri Sembilan

NAMA PELAJAR	: AIMAN AYER
FACULTY	: FACULTY SCIENCE INFROMATION
PROGRAM	: SARJANA MUDA SAINS MAKLUMAT (KEPUJIAN) PENGURUSAN SISTEM MAKLUMAT
EMAIL	: AIMAN@AYER.COM
PHONE	: 0112223333
GROUP MEMBER	: 4

PERMOHONAN Sesi TEMUBUAL UNTUK MENJALANKAN KAJIAN
 Dengan segala hormatnya saya merujuk kepada perkara di atas.

2. Suka-cita Dimaklumkan bahawa kami seramai 4 orang Pelajar daripada Program Kode CDIM262 **SARJANA MUDA SAINS MAKLUMAT (KEPUJIAN) PENGURUSAN SISTEM MAKLUMAT** ingin Mejalankan satu kajian dalam bentuk Sesi TemuBual di **Abc Sdn.Bhd** secara Online bagi menyiapkan Tugas yang telah Diberi.Pada tarikh **08 June 2026**(tertakluk kepada perubahan dan cuti mingguan organisasi).

3. Di samping itu, kami memohon jasa baik pihak Tuan/puan untuk membenarkan kami merakam sesi temubual tersebut bagi tujuan pembelajaran berterusan. Dengan ini kami sertakan maklumat untuk pihak puan hubungi Aiman ayer (aiman@ayer.com dan 0112223333)

4. Pertimbangan dan kerjasama pihak Tuan/Puan dalam perkara ini amatlah dihargai. Bersama-sama ini disertakan surat sokongan untuk makluman

Application Data

Approval Date	Jun 21 2026(07:50 PM)
Approval date	Jul 04 2026(05:39 PM)
Application Status	Approve

Approval Status

approval

[Download PDF](#)
[Reset](#)
[Submit](#)

Picture 2.

The system shows a detailed view interface to provide a detailed preview of the official letter created as well as its administrative management controls in a 2-column layout. The left column shows the dynamically generated letter, including the institutional logo of Universiti Teknologi MARA (UiTM), the formal reference number, and company recipient information and a detailed breakdown of the student group's metadata, including the student leader's name, program, and group size. The interface displays the official letter body in Malay (Permohonan Sesi Temubual Untuk Menjalankan Kajian) below the student particulars section, which includes the academic reason, course code and timetable information of the interview. The "Application Data" control panel is in the right column, and it enables administrators to monitor audit trail data such as the date of the first approval and when the application was last updated and shows a visual "Approve" status indicator badge. This panel lets administrators alter the status of a document using a pulldown menu or perform resets or upload the status of the document to the system database directly from the panel through action utility buttons.